Implementation Plan: Knowledgebase Assistant Workflow (Topic 11)

1. Document Overview

This document describes the implementation of the Governed Knowledgebase Assistant Workflow using Google Opal.

The workflow classifies employee queries, routes operational questions to a grounded research agent, and prepares escalation notifications for sensitive HR, legal, or compliance matters.

2. Configuration Flow

The implemented workflow follows this sequence:

1. Employee Query – collects the employee's workplace question.
2. Classification Result – classifies the query as OPERATIONAL or SENSITIVE.
3. Governance Classifier – routes the query to the appropriate workflow agent.
4. Grounded Research Agent – generates operational answers using uploaded SOP assets.
5. Human Escalation Handler – prepares a human-review notification for sensitive queries.
6. Display response – displays the appropriate workflow response.

3. Step-by-Step Implementation

Step 1: Create the Workflow

- Create a Google Opal application named Knowledgebase Assistant.
- Configure the employee query input.
- Provide a clear description of the assistant's purpose and limitations.

Step 2: Upload SOP Assets

Upload the following fictional demonstration SOP documents:

- `employee_leave_policy.md`
- `it_support_sop.md`
- `expense_reimbursement_policy.md`

These documents provide the static knowledge sources for operational responses.

Important: These are fictional training documents and must be replaced with approved company policies before real workplace use.

Step 3: Configure Employee Query

Create an input node named Employee Query.

Configure it to accept workplace questions about company policies, procedures, IT support, leave requests, and expense reimbursement.

Tell employees not to enter passwords, OTPs, or confidential credentials.

Step 4: Configure Classification

Create a classification step named Classification Result.

Configure it to return exactly one of these classifications:

- OPERATIONAL
- SENSITIVE

Operational examples include routine leave requests, IT support, software access, and expense reimbursement.

Sensitive examples include harassment, discrimination, retaliation, termination disputes, salary grievances, legal threats, fraud, and contract interpretation.

Ambiguous or mixed queries that may involve sensitive matters should be classified as SENSITIVE.

Step 5: Configure Governance Routing

Create an agent named Governance Classifier.

Connect it to the Employee Query and Classification Result inputs.

Configure two routing actions:

- OPERATIONAL → Go to Grounded Research Agent.
- SENSITIVE → Go to Human Escalation Handler.

The workflow must execute only the appropriate routing action.

Step 6: Configure Grounded Research Agent

Create an agent named Grounded Research Agent.

Connect the employee query, classification result, and three uploaded SOP assets.

Configure the agent to:

1. Answer operational questions only.
2. Retrieve relevant information from the uploaded SOP documents.
3. Generate answers using only documented information.
4. Identify the source SOP document.
5. State clearly when the available documents do not contain the requested information.
6. Avoid inventing policies, deadlines, approval limits, or employee entitlements.
7. Avoid providing legal advice or interpreting employment contracts.

Step 7: Configure Human Escalation Handler

Create an agent named Human Escalation Handler.

Connect the employee query and classification result.

Configure it to prepare a structured escalation notification for SENSITIVE queries.

Recipient: hr_manager@company.com

Subject: URGENT: Human Review Required – Sensitive Employee Query

The notification should contain:

- Original employee query.
- Employee details, if provided.
- Classification.
- Reason for escalation.
- Required human action.

If employee details are unavailable, display Not Provided.

The handler must not independently investigate or resolve the complaint.

Implementation limitation: The escalation notification is prepared as workflow output. Actual email delivery is not configured or confirmed.

Step 8: Configure Display Response

Create a render-output step named Display response.

Configure it to display:

- The Grounded Research Agent response when classification is OPERATIONAL.
- The Human Escalation Handler notification when classification is SENSITIVE.

Do not display the unselected agent's output.

Preserve the query, classification, answer or escalation details, and source where applicable.

Step 9: Preview and Test

Run the workflow preview with operational and sensitive employee queries.

Verify:

- Operational classification.
- Sensitive classification.
- Correct routing.
- SOP-grounded answer generation.
- Escalation notification preparation.
- Appropriate final response display.

Inspect the execution details and console for errors where available.

Refer to `testing_plan.md` for the recorded test cases and results.

Step 10: Theme and Publishing

- Review the application interface and theme.
- Check the final preview.
- Publish the application only after reviewing the workflow and its limitations.
- Configure sharing permissions according to the intended demonstration or access requirements.

Publishing and sharing must be verified separately.

4. Implementation Status

Component	Status
Employee Query	Implemented
Classification Result	Implemented
Governance Classifier	Implemented
Grounded Research Agent	Implemented

Human Escalation Handler	Implemented
Display response	Configured; conditional display must be verified
SOP assets	Uploaded
Email delivery	Not configured
Preview testing	Completed for recorded scenarios
Publishing	Verified

5. Implementation Conclusion

The Knowledgebase Assistant workflow demonstrates governed query classification, operational knowledge retrieval, and sensitive-query escalation notification preparation.

The workflow is designed to use uploaded SOP documents as its knowledge sources and to refer sensitive matters for human review.

Actual email delivery and production use with approved company policies require additional configuration and verification.